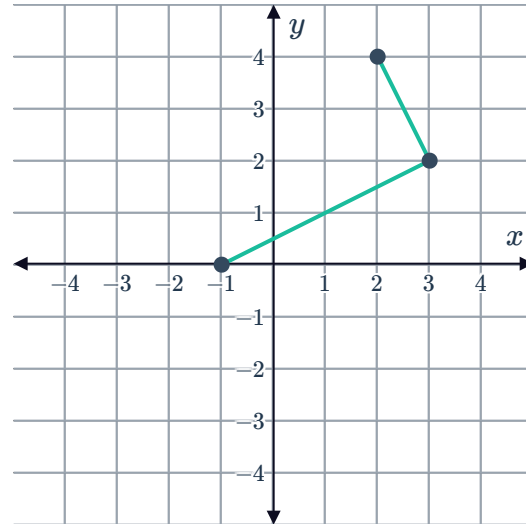


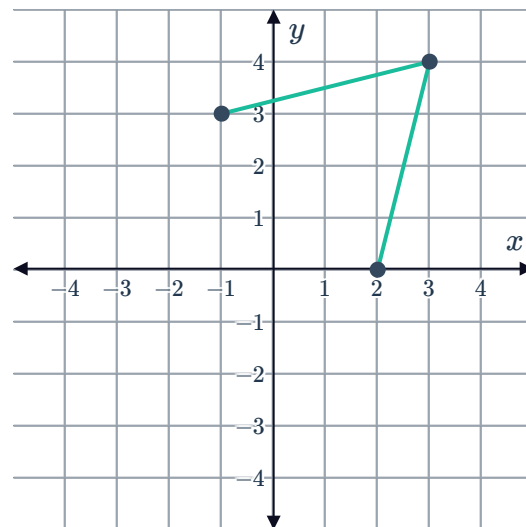
## Coordinate methods with quadrilaterals



- 1 Consider the given diagram.  
The points given represent three vertices of a parallelogram.  
Determine the coordinates of the fourth vertex.

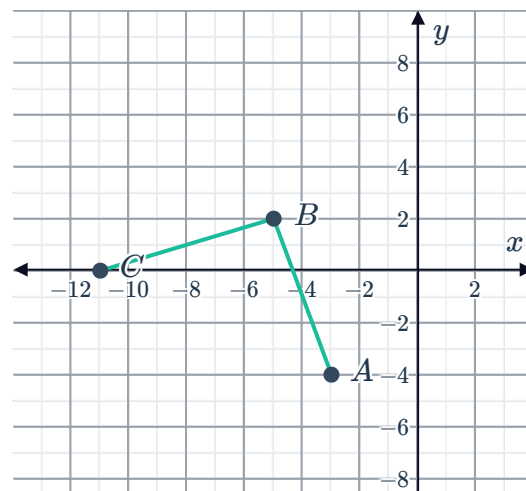


- 2 Consider the given diagram.  
The points given represent three vertices of a rhombus.  
Determine the coordinates of the fourth vertex if the missing point lies in the 3rd quadrant.



- 3 Consider the given diagram.  
 $A(-3, -4)$ ,  $B(-5, 2)$ ,  $C(-11, 0)$ , and  $D(p, q)$  form the four vertices of square  $ABCD$ , where  $T$  is the midpoint of the diagonal  $\overline{AC}$ .

- a Find the coordinates of  $T$ .
- b Solve for  $p$ .
- c Solve for  $q$ .





4 Consider the quadrilateral shown.

a Find the slope of the following:

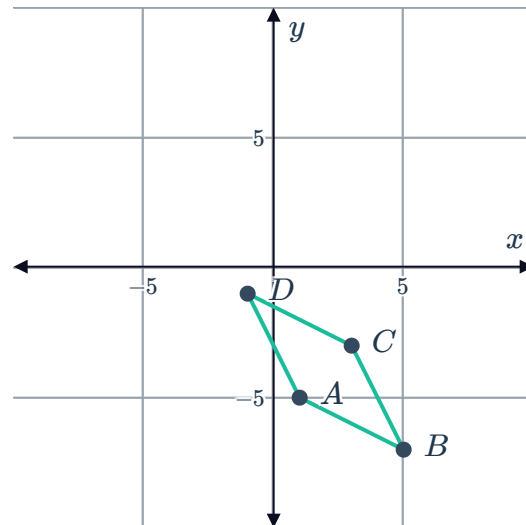
i  $\overline{AD}$                       ii  $\overline{AB}$

iii  $\overline{BC}$                       iv  $\overline{DC}$

b Find the length of the following sides:

i  $\overline{AB}$                       ii  $\overline{AD}$

c Is this quadrilateral a rhombus or just a parallelogram?



5 The four points  $A(-8, -4)$ ,  $B(-5, 0)$ ,  $C(0, -2)$  and  $D(-3, -6)$  are the vertices of a quadrilateral.

a Plot the quadrilateral on a coordinate plane.

b Find the slope of the following:

i  $\overline{AB}$                       ii  $\overline{BC}$                       iii  $\overline{CD}$                       iv  $\overline{DA}$

c Write down two pairs of segments that are parallel.

d Hence, classify the quadrilateral as precisely as possible.

6 Consider the quadrilateral shown.

a Find the slope of the following sides:

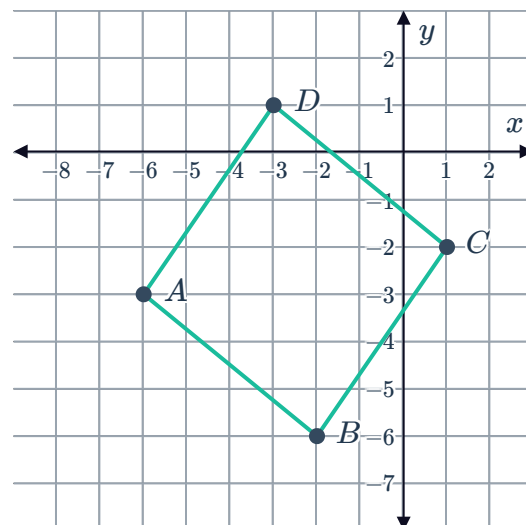
i  $\overline{AB}$                       ii  $\overline{CD}$

iii  $\overline{AD}$                       iv  $\overline{BC}$

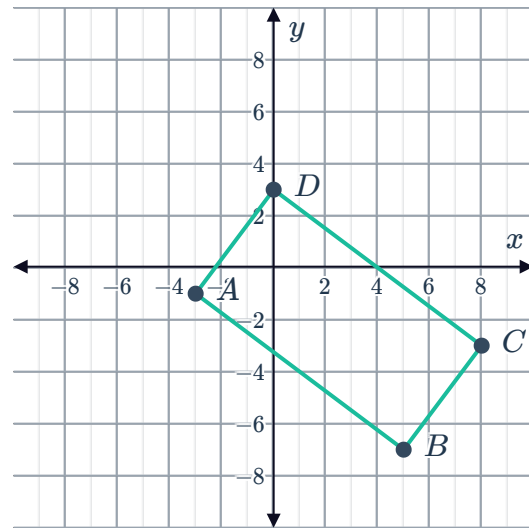
b Find the length of the following sides:

i  $\overline{AB}$                       ii  $\overline{AD}$

c Hence, classify the quadrilateral as precisely as possible.



7 Consider the quadrilateral shown.



a Find the slope of the following sides:

- i  $\overline{AB}$                       ii  $\overline{CD}$   
iii  $\overline{AD}$                       iv  $\overline{BC}$

b Find the length of the following sides:

- i  $\overline{AB}$                       ii  $\overline{AD}$

c Hence, classify the quadrilateral as precisely as possible.

## Additional questions

8 State the properties of each of the following triangles that can be identified with coordinate geometry:

- a Scalene triangles                      b Isosceles triangles  
c Equilateral triangles                      d Right triangles

9 Consider the quadrilateral formed by the points  $A(-3, -4)$ ,  $B(5, -4)$ ,  $C(5, 4)$  and  $D(-3, 4)$ :

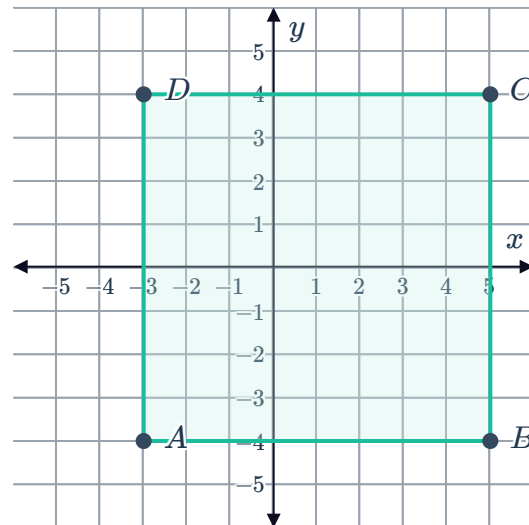
a Find the length of the following sides:

- i  $\overline{AB}$     ii  $\overline{BC}$     iii  $\overline{CD}$     iv  $\overline{DA}$

b State if all sides of the quadrilateral are congruent.

c State if all consecutive sides are perpendicular.

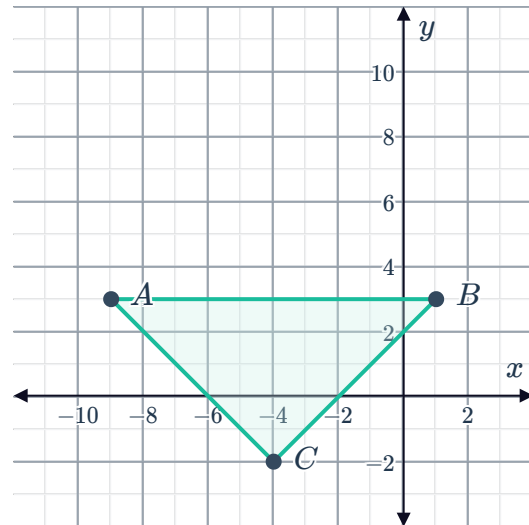
d Now, classify the quadrilateral as precisely as possible.



10 Consider the triangle shown below:

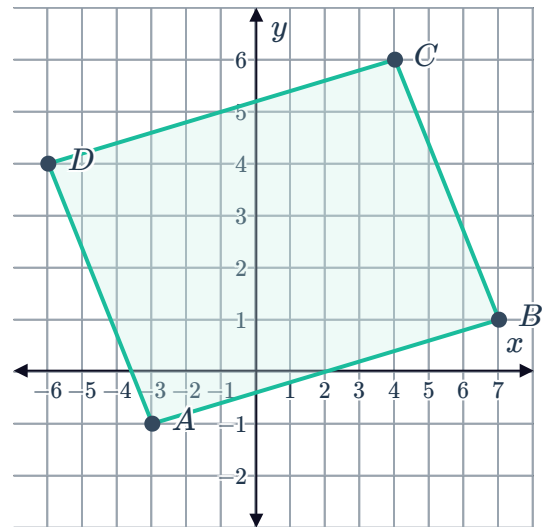


- a Find the slope of the following sides:
  - i  $\overline{AC}$
  - ii  $\overline{BC}$
- b Calculate the side lengths of all three sides.
- c Classify the triangle as precisely as possible.



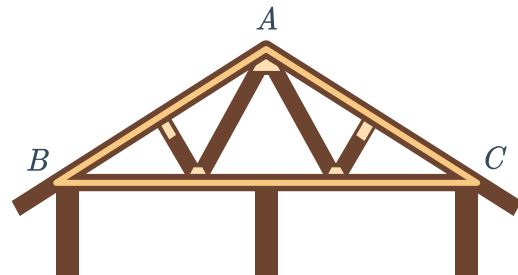
11 Consider the parallelogram  $ABCD$ :

- a Find the slope of the following sides:
  - i  $\overline{AB}$
  - ii  $\overline{CD}$
  - iii  $\overline{AD}$
  - iv  $\overline{BC}$
- b Classify this quadrilateral as precisely as possible.

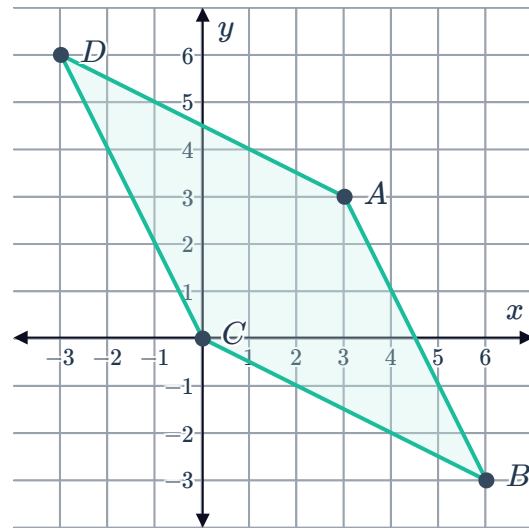


12 Isolda is designing a house using a graphing software. The outer pieces of the triangular roof trusses must form an isosceles triangle. The outer pieces of one of the triangular trusses is formed by three points:  $A(2, 2)$ ,  $B(-2, 0)$  and  $C(6, 0)$ .

- a Calculate the following lengths:
  - i  $BC$
  - ii  $AB$
  - iii  $AC$
- b State if this is a possible roof truss.



- 13 Consider the quadrilateral  $ABCD$ :  
Classify this quadrilateral as precisely as possible.



- 14 Consider a quadrilateral with vertices  $A(3, 4)$ ,  $B(6, 2)$ ,  $C(5, -3)$ , and  $D(-4, 3)$ :
- Plot the quadrilateral on a coordinate plane.
  - Classify the quadrilateral  $ABCD$  as precisely as possible.
- 15 Consider a quadrilateral with vertices  $P(4, -4)$ ,  $Q(5, -7)$ ,  $R(9, -11)$  and  $S(0, -8)$ .  
Classify the quadrilateral  $PQRS$  as precisely as possible.
- 16 A triangle consists of the points  $A(-2, -2)$ ,  $B(1, -4)$  and  $C(5, 2)$ .  
Classify  $\triangle ABC$  as precisely as possible.

- 17 Consider the following lines:

- Line P:  $y = -6x - 4$
- Line Q:  $y = \frac{x}{6} + 6$

- Line R:  $y = -6x - 1$
- Line S:  $y = \frac{x}{6} + 1$

- a Find the slope of each line:

i  $m_P$

ii  $m_Q$

iii  $m_R$

iv  $m_S$

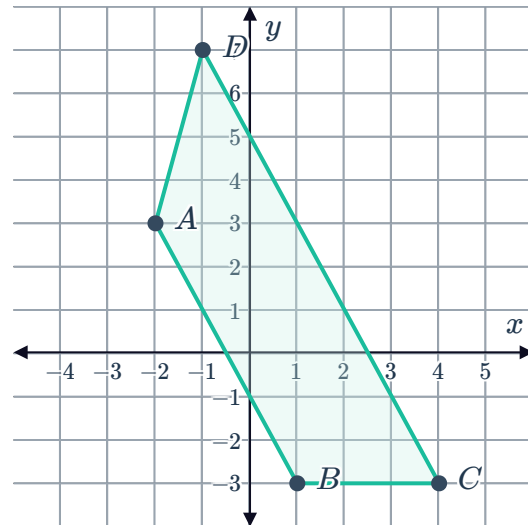
- b State the type of the quadrilateral enclosed by the four lines.

- 18 Quadrilateral  $ABCD$  has vertices  $A(1, -2)$ ,  $B(9, 3)$ ,  $C(7, -1)$  and  $D(-1, -6)$ :  
Prove that  $ABCD$  the diagonals bisect each other and therefore that  $ABCD$  is a parallelogram.

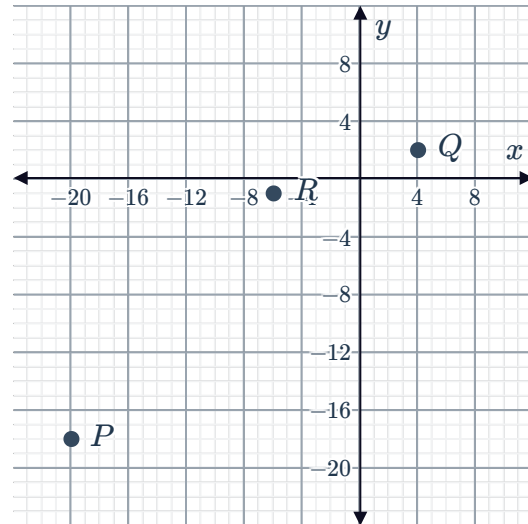
- 19 Quadrilateral  $RSTU$  has vertices  $R(3, 0)$ ,  $S(5, -3)$ ,  $T(-1, -7)$ ,  $U(-3, -4)$ .



- 20** Prove the the given quadrilateral is a trapezoid, but not an isosceles trapezoid.



- 21** Consider the points  $P(-20, -18)$ ,  $Q(4, 2)$  and  $R(-6, -1)$ :  
Determine the coordinates of a point  $S$   
such that  $PQRS$  is a parallelogram.



- 22** Describe two ways you can show whether a quadrilateral in the coordinate plane is a square.  
Explain which method is more efficient.